KONGU ENGINEERING COLLEGE

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DEPARMENT OF CIVIL ENGINEERING

SURVEY CAMP - ONE CREDIT COURSE

THIRD YEAR STUDENTS

ACADEMIC YEAR 2025 - 2026

FEBRUARY - 2026

KONGU

Assuring the best

BONAFIDE CERTIFICATE

Name :SELVAM P S K

Roll No. : 23CER044

Programme : B.E.

Branch : Civil Engineering

Section : -

Semester : VI

Course Code : 22VC130

Course Name : Survey Camp

Certified that this is a bonafide record of work done by the above student during the academic year 2025 - 2026.

Signature of Faculty in-charge Signature of Head of the Department

Submitted for the practical examination held on _____.

Examiner – I Examiner – II

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Average (60):

Faculty Sign:

EXPERIMENT 1

Longitudinal and Cross-Section Survey for Highway Project Using Dumpy Level

Aim

To conduct longitudinal and cross-section levelling along a proposed highway alignment using a dumpy level and to determine the ground profile.

Apparatus / Equipment

Dumpy level, tripod stand, leveling staff, measuring tape, chain, ranging rods, wooden pegs, hammer, field book, pencil.

Detailed Procedure

The survey work is commenced with a reconnaissance of the proposed highway alignment to understand the nature of the terrain and to select suitable positions for instrument setup. The center line of the highway is marked on the ground using ranging rods. Chainages are fixed along the center line at regular intervals, generally at 10 m or 20 m spacing.

A benchmark with known reduced level is identified near the starting point of the alignment. The dumpy level is set up on firm ground and carefully leveled by adjusting the leveling screws until the bubble remains centered. A back sight reading is taken on the benchmark using a leveling staff held vertically. Intermediate sight readings are then taken at each chainage point along the alignment.

When the instrument position needs to be changed, a fore sight reading is taken on a change point. The instrument is shifted to a new position and a back sight reading is again taken on the same change point to continue the levelling work.

For cross-section surveying, offsets are measured perpendicular to the center line on both sides at selected chainages. Staff readings are taken at each offset point. All observations are recorded systematically in the level field book. Reduced levels are calculated using the height of instrument method and longitudinal and cross-section profiles are plotted.

Tabulation|

Chainage	BS(m)	IS(m)	FS(m)	HI(m)	RL(m)	Remarks
	0.08			100.08	100	BM
0		0.345			99.74	
		0.365			99.72	
		0.385			99.70	
10		0.655			99.43	
		0.715			99.37	
		0.685			99.40	
20		1.105			98.98	
		1.085			99.00	
		1.065			99.02	
30		1.365			98.72	
		1.400			98.68	
		1.385			98.70	
40		1.775			98.31	
		1.815			98.27	
		1.800			98.28	
50		2.215			97.87	
		2.175			97.91	
		2.125			97.96	
60		2.615			97.47	
		2.665			97.42	
		2.625			97.46	
70		3.195			96.89	
		3.135			96.95	
		3.105			96.98	
80		3.645			96.44	
		3.600			96.48	

90	0.16		3.720	96.52	96.36	CP1
		0.715			95.81	
		0.815			95.71	
		0.655			95.87	
100		0.385			96.14	
		0.485			96.04	
		0.425			96.10	
110		0.268			96.25	
		0.235			96.29	
		0.285			96.24	
120		2.065			94.46	
		1.965			94.56	
		2.115			94.40	
130		1.765			94.76	
		1.770			94.75	
		1.800			94.72	
140		1.485			95.04	
		1.435			95.09	
		1.530			94.99	
150		1.175			95.35	
		1.115			95.41	
		0.245			96.28	
160		0.875			95.65	
		0.855			95.67	
		0.920			95.60	
170		0.555			95.97	
		0.545			95.98	
	0.175		0.605	95.75	95.92	CP2
180		0.195			95.55	

		0.165			95.59	
		0.125			95.63	
190		0.845			94.91	
		0.825			94.93	
		0.905			94.85	
200		0.485			95.27	
		0.415			95.34	
		0.525			95.22	

Calculations

Height of Instrument (HI) = RL of BM + BS
RL = HI − Staff Reading

Result

The longitudinal and cross-section profiles of the proposed highway alignment were successfully determined.

EXPERIMENT 2

Longitudinal and Cross-Section Survey for Water Supply Project

Aim

To determine longitudinal and cross-section levels along a proposed water supply pipeline or canal alignment.

Apparatus / Equipment

Dumpy level, tripod, leveling staff, measuring tape, ranging rods, pegs, field book.

Detailed Procedure

The alignment of the proposed water supply project is identified and marked on the ground using ranging rods. Chainages are fixed at uniform intervals along the alignment. A suitable benchmark is selected near the starting point.

The dumpy level is set up and leveled properly. A back sight reading is taken on the benchmark to establish the height of instrument. Intermediate sight readings are taken at all chainage points. Change points are used whenever the instrument is shifted.

At selected chainages, cross-sections are taken by measuring offsets on both sides of the alignment. Staff readings are recorded at all offset points. Reduced levels are computed and longitudinal and cross-section drawings are prepared to assess excavation depth and gradient.

Tabulation

Chainage	BS(m)	IS(m)	FS(m)	HI(m)	RL(m)	Remarks
	0.08			100.08	100	BM
0		0.345			99.74	
		0.365			99.72	
		0.385			99.70	
10		0.655			99.43	
		0.715			99.37	
		0.685			99.40	
20		1.105			98.98	
		1.085			99.00	
		1.065			99.02	
30		1.365			98.72	
		1.400			98.68	

		1.385			98.70	
40		1.775			98.31	
		1.815			98.27	
		1.800			98.28	
50		2.215			97.87	
		2.175			97.91	
		2.125			97.96	
60		2.615			97.47	
		2.665			97.42	
		2.625			97.46	
70		3.195			96.89	
		3.135			96.95	
		3.105			96.98	
80		3.645			96.44	
		3.600			96.48	
90	0.16		3.720	96.52	96.36	CP1
		0.715			95.81	
		0.815			95.71	
		0.655			95.87	
100		0.385			96.14	
		0.485			96.04	
		0.425			96.10	
110		0.268			96.25	
		0.235			96.29	
		0.285			96.24	
120		2.065			94.46	
		1.965			94.56	
		2.115			94.40	
130		1.765			94.76	

		1.770			94.75	
		1.800			94.72	
140		1.485			95.04	
		1.435			95.09	
		1.530			94.99	
150		1.175			95.35	
		1.115			95.41	
		0.245			96.28	
160		0.875			95.65	
		0.855			95.67	
		0.920			95.60	
170		0.555			95.97	
		0.545			95.98	
180	0.175		0.605	95.75	95.92	CP2
		0.195			95.55	
		0.165			95.59	
		0.125			95.63	
190		0.845			94.91	
		0.825			94.93	
		0.905			94.85	
200		0.485			95.27	
		0.415			95.34	
		0.525			95.22	

Calculations

Same as Experiment 1.

Result

The required ground profile for the water supply project was successfully obtained.

EXPERIMENT 3

Preparation of Contour Map Using Total Station

Aim

To prepare a contour map of a given area using total station survey.

Apparatus / Equipment

Total station, tripod, prism with pole, data cable, computer, AutoCAD software.

Detailed Procedure

The survey area is inspected and suitable control points are identified. The total station is set up over a known point and accurately centered using optical or laser plummet. The instrument is leveled carefully and oriented by taking a backsight to a known reference point.

Spot heights are observed systematically over the entire area to capture variations in ground elevation. At each point, horizontal angle, vertical angle, and slope distance are measured and coordinates and elevations are recorded. The collected data is transferred to a computer and imported into AutoCAD software. Contours are generated at the required interval by interpolation.

Tabulation

Point No.	Easting (m)	Northing (m)	Elevation (m)
1	0.00	0.00	0.00
2	978.184	984.533	106.081
3	978.259	987.410	104.940
4	978.204	991.022	103.936
5	979.052	994.652	102.934
6	981.537	996.242	102.151
7	979.058	999.589	101.224
8	980.174	1003.376	100.107
9	976.100	1007.874	98.521
10	979.269	1009.317	98.044

11	981.696	1010.144	97.779
12	985.654	1015.870	95.672
13	988.061	1019.592	94.532
14	990.885	1022.243	94.163
15	994.801	1020.987	93.839
16	994.155	1018.031	94.777
17	992.880	1015.291	95.715
18	993.254	1012.495	96.533
19	993.352	1009.664	97.392
20	992.992	1006.404	98.693
21	993.788	1003.479	99.436
22	992.386	1001.011	99.925
23	991.158	999.717	100.379
24	991.356	996.690	100.653
25	991.443	995.131	101.807
26	987.587	991.791	102.849
27	986.669	989.479	103.580
28	986.836	986.269	104.283
29	984.783	979.913	107.224
30	990.103	978.930	106.897
31	994.677	978.121	106.437
32	997.007	980.323	105.361
33	999.118	980.158	105.375
34	1000.868	979.848	105.647

35	1006.973	979.282	103.955
36	1011.184	981.054	102.762
37	1007.511	983.402	102.058
38	1003.814	985.303	101.652
39	1000.630	986.840	101.729
40	997.155	988.853	102.057
41	994.106	990.289	102.401
42	992.251	992.829	102.341
43	995.690	994.077	101.176
44	996.333	995.697	100.644
45	992.682	997.956	100.412
46	992.028	1000.764	99.992
47	992.462	1003.263	99.383
48	993.001	1005.078	99.009
49	995.842	1003.650	99.376
50	997.501	1002.366	99.611
51	998.825	1001.473	99.810
52	998.131	999.192	100.022
53	998.461	997.184	100.292
54	999.118	995.616	100.562
55	999.972	993.892	100.752
56	1000.939	992.043	100.995
57	1002.590	991.049	100.977
58	1004.718	989.701	101.134

59	1007.192	988.569	101.115
60	1008.847	989.499	100.951
61	1010.983	987.148	101.289
62	1011.089	989.741	100.895
63	1008.642	991.662	100.658
64	1006.762	993.348	100.454
65	1004.033	995.982	100.277
66	1001.989	997.785	100.130
67	999.738	997.429	100.214
68	1003.539	1001.700	99.476
69	1004.806	1000.394	99.590
70	1005.711	998.970	99.217
71	1007.787	997.904	98.852
72	1007.781	996.393	99.317
73	1008.403	994.547	99.936
74	1010.576	995.026	99.096
75	1011.396	996.370	98.685
76	1013.858	994.565	98.744
77	1013.864	994.572	98.746
78	1016.322	993.056	98.660
79	1015.741	991.025	99.431
80	1015.248	988.881	100.185
81	1014.670	986.328	101.197
82	1013.926	985.112	101.941

83	1017.623	984.029	100.929
84	1019.311	985.801	100.203
85	1021.130	987.506	99.311
86	1022.161	989.089	98.598
87	1021.145	991.685	98.125
88	1017.550	994.361	98.126
89	1016.172	992.202	98.902
90	1015.357	998.283	97.477
91	1015.697	1001.323	96.592
92	1016.919	1003.943	95.667
93	1018.716	1006.480	94.627
94	1019.496	1008.401	93.967
95	1017.354	1011.098	93.735
96	1015.530	1012.685	93.674
97	1013.513	1009.940	94.802
98	1011.741	1006.675	95.924
99	1010.177	1003.945	96.894
100	1009.176	1001.773	97.566
101	1006.187	1001.261	98.571
102	1007.257	1003.359	97.530
103	1005.569	1004.320	97.624
104	1003.735	1005.004	97.652
105	1001.709	1005.776	97.726
106	1000.813	1006.961	97.521

107	1001.860	1008.989	96.843
108	1004.440	1008.450	96.686
109	1005.555	1010.404	95.960
110	1006.900	1012.399	95.248
111	1004.406	1013.476	95.332
112	1002.682	1011.697	96.029
113	1001.236	1010.756	96.432
114	999.124	1011.140	96.531
115	999.520	1013.144	95.854
116	1000.438	1015.819	94.930
117	1002.962	1014.523	95.106
118	998.878	1016.914	94.752
119	998.517	1014.108	95.639
120	997.885	1012.063	96.381
121	996.573	1008.942	97.405
122	995.046	1006.915	98.177
123	993.886	1009.347	97.383
124	994.230	1013.023	96.294
125	995.437	1015.294	95.539
126	995.461	1018.447	94.522
127	993.524	1019.509	94.365
128	992.482	1015.507	95.711
129	991.393	1009.763	97.454
130	1018.254	984.717	100.672

131	1016.687	981.293	101.748
132	1015.778	977.382	103.023
133	1014.372	973.431	104.301
134	1017.036	971.841	104.379
135	1019.088	975.152	103.157
136	1020.899	976.807	102.259
137	1022.816	982.169	100.515
138	1023.913	985.071	99.567
139	1027.428	982.974	99.398

Calculations

Coordinates and elevations are computed automatically by the total station.

Result

The contour map of the given area was prepared successfully.

EXPERIMENT 4

Estimation of Area of Irregular Boundary Using Total Station

Aim

To determine the area of an irregular field using total station survey.

Apparatus / Equipment

Total station, tripod, prism, data cable, AutoCAD software.

Detailed Procedure

The survey work is started with a reconnaissance of the given irregular field to understand the shape of the boundary, visibility between points, and suitability of instrument stations. All boundary corner points of the field are identified clearly on the ground and temporarily marked using wooden pegs or paint so that they can be observed easily during the survey.

A suitable station point is selected at a location from where maximum boundary points are visible. The total station is set up firmly on the tripod over the selected station point. The instrument is accurately centered over the station point using the optical or

laser plummet and then carefully leveled by adjusting the foot screws until the circular and tubular bubbles are perfectly centered.

The total station is oriented by taking a backsight to a known reference point or by setting an assumed azimuth. After orientation, the prism mounted on a prism pole is placed vertically over the first boundary corner point. The horizontal angle, vertical angle, and slope distance to the prism are observed, and the corresponding Easting and Northing coordinates of the point are recorded automatically by the total station.

The prism is then shifted sequentially to each boundary corner point of the irregular field, and observations are taken in the same manner. Care is taken to observe the boundary points in proper sequence (clockwise or anticlockwise) so that a closed traverse of the field is obtained. All observed coordinates are stored in the total station memory.

After completing the field observations, the recorded data is transferred from the total station to a computer using a data cable or USB device. The data file is imported into AutoCAD software. Using the coordinate data, all boundary points are plotted accurately in AutoCAD and joined in sequence to form a closed polygon representing the field boundary.

Once the closed boundary is verified for correctness, the area enclosed by the polygon is calculated using the AREA command in AutoCAD. The calculated area is noted and converted into required units if necessary. The plotted field boundary and area calculation are checked to ensure accuracy.

Tabulation

Point No	Easting(m)	Northing(m)
1	0.000	0.000
2	1023.624	100.483
3	1020.433	100.476
4	963.385	98.851
5	968.076	98.766
6	970.921	99.088
7	970.749	98.959
8	972.825	99.108
9	977.147	99.276

10	986.117	99.461
11	994.125	99.621
12	1002.304	99.738
13	1012.072	99.755
14	1019.94	99.726
15	1028.369	99.535
16	1035.562	99.333
17	1043.615	99.04
18	1051.31	98.748
19	1056.544	98.528
20	1069.801	98.005
21	1076.746	97.64
22	1090.543	97.107
23	1103.889	96.568
24	1114.26	96.301
25	1121.567	96.54
26	1127.165	96.766
27	1126.94	96.904
28	1131.717	96.937
29	1134.849	97.011
30	1140.275	96.287
31	1137.241	96.863
32	1137.753	96.942
33	1137.694	96.94

34	1137.421	97.012
35	1135.567	97.025
36	1131.051	96.884
37	1126.904	96.77
38	1122.064	96.854
39	1120.437	96.974
40	1120.251	97.023
41	1120.268	97.174
42	1120.572	97.243
43	1120.49	97.302
44	1119.182	97.503
45	1119.171	97.621
46	1119.554	97.667
47	1124.438	97.82
48	1121.952	97.727
49	1120.569	97.822
50	1120.181	97.928
51	1120.148	97.978
52	1120.477	98.026
53	1121.702	98.191
54	1122.064	98.204
55	1122.842	98.192
56	1123.654	98.203
57	1124.204	98.205

58	1124.784	98.191
59	1125.519	98.19
60	1125.63	98.179
61	1126.305	98.18
62	1126.597	98.186
63	1126.872	98.165
64	1126.906	98.167
65	1126.476	98.192
66	1126.401	98.194
67	1126.306	98.191
68	1128.491	98.471
69	1128.19	98.493
70	1128.422	98.407
71	1129.604	98.318
72	1119.837	98.429
73	1108.115	98.399
74	1099.447	98.848
75	1093.903	99.135
76	1090.149	99.171
77	1086.094	99.274
78	1080.156	99.378
79	1076.991	99.462
80	1074.994	99.385
81	1070.533	99.632

82	1062.429	99.887
83	1054.755	100.077
84	1040.256	100.315
85	1033.593	100.317
86	1030.042	100.619
87	1026.125	100.354
88	1012.444	99.925
89	1007.983	99.864
90	1001.195	99.488
91	994.79	99.48
92	988.408	99.387
93	983.234	99.58
94	980.144	99.54
95	968.888	99.679
96	965.854	99.706
97	960.844	99.721
98	953.621	99.747
99	942.935	99.551
100	931.853	99.189
101	931.312	99.167
102	928.83	98.969
103	928.092	99.067
104	927.534	99.041
105	929.502	99.127

106	927.944	98.997
107	926.514	98.866
108	925.438	98.751
109	923.116	98.719
110	922.4	98.563
111	925.956	98.764
112	930.634	98.924
113	938.22	99.199
114	942.461	99.157
115	950.749	99.008
116	953.996	98.966
117	958.976	98.938
118	962.991	98.868
119	965.721	98.893
120	965.945	98.861
121	966.273	98.581
122	966.479	98.589
123	1023.624	100.483
124	1020.433	100.476
125	963.385	98.851
126	968.076	98.766
127	970.921	99.088
128	970.749	98.959
129	972.825	99.108

Calculations

Area is computed using AutoCAD or coordinate geometry.

Result

The area of the irregular field was accurately determined.

EXPERIMENT 5

Marking of Building Boundaries Using Total Station

Aim

To set out and mark the building boundary points accurately on the ground using a total station based on given design coordinates or plan dimensions.

Apparatus / Equipment

Total station, tripod stand, prism with prism pole, measuring tape, wooden pegs, hammer, lime powder or chalk, field notebook.

Detailed Procedure

The work is started by studying the building plan or drawing to understand the dimensions, orientation, and coordinates of the building boundaries. Suitable reference control points are identified or established near the site to serve as base points for setting out the building. These control points should be stable and clearly visible from the proposed building location.

The total station is set up firmly on the tripod over one of the selected control points. The instrument is accurately centered using the optical or laser plummet and then carefully leveled by adjusting the foot screws until the circular and tubular bubbles are centered. The total station is oriented by taking a backsight observation to another known control point so that the instrument is properly aligned with the site reference system.

The design coordinates of the building corner points are then entered into the total station or data collector. The instrument is set to stake-out mode. The prism mounted on a prism pole is moved on the ground as guided by the total station display until the exact position of the first building corner is located. When the prism reaches the correct design position within the permissible tolerance, the point is marked on the ground using a wooden peg.

The same procedure is repeated for all remaining building corner points. Each point is carefully marked and labeled to avoid confusion. After marking all corners, the boundary lines of the building are checked by measuring the distances between adjacent corners and by checking the diagonals to ensure accuracy and squareness of the layout.

Finally, lime powder or chalk is used to clearly mark the building boundary lines by joining the marked corner points. The entire layout is rechecked to confirm that the building boundaries have been set out correctly as per the given plan.

Tabulation

Corner No	Design Easting(m)	Design Northing(m)
A1	1872.01	1695.18
A2	1872.01	1684.64
A3	1872.01	1675.21
A4	1872.01	1665.55
B1	1883.56	1695.18
B2	1883.56	1684.90
B3	1883.56	1675.21
C1	1891.42	1695.18
C2	1891.42	1684.64
C3	1891.42	1665.55

Calculations

Coordinate differences and deviations between design points and field-marked points are computed automatically by the total station during the stake-out process. Linear measurements and diagonal checks are used to verify accuracy.

Result

The building boundary points were successfully set out and accurately marked on the ground using the total station as per the given design plan.

EXPERIMENT 6

Determination of Area of Irregular Field Using DGPS

Aim

To determine the area of an irregular field accurately using Differential Global Positioning System (DGPS).

Apparatus / Equipment

DGPS base receiver, DGPS rover receiver, tripod, controller/data logger, batteries, communication cables, DGPS processing software, field notebook.

Detailed Procedure

The survey work is started with a reconnaissance of the irregular field to understand the boundary shape, accessibility, and satellite visibility conditions. All corner points of the field boundary are identified clearly and temporarily marked so that they can be easily observed during the survey.

The DGPS base station is first set up over a known reference point with established coordinates. The base receiver is mounted on a tripod and accurately centered over the point using the optical or laser plummet. The antenna height is measured and recorded carefully. The base station is then powered on and allowed to initialize until it starts transmitting correction signals.

After the base station is initialized, the DGPS rover receiver is switched on and connected to the controller. The rover is allowed to initialize and receive real-time correction signals from the base station. Proper satellite lock and acceptable accuracy levels are ensured before commencing the survey.

The rover receiver is then carried to the first boundary corner point of the irregular field. The rover pole is held vertically over the point and observations are recorded once the position accuracy stabilizes. Latitude, longitude, and elevation of the point are stored in the controller memory. The rover is moved sequentially to all boundary corner points, and coordinates are recorded in proper order so as to form a closed boundary of the field.

After completing the field observations, the collected DGPS data is transferred to a computer using appropriate data transfer software. The recorded coordinates are processed to apply differential corrections and improve positional accuracy. The corrected boundary points are plotted, and a closed polygon representing the field boundary is formed. The area enclosed by the polygon is computed automatically by the DGPS processing software.

The computed area is checked for consistency and converted into required units if necessary. All results are documented systematically.

Tabulation

Point No	Latitude	Longitude	Elevation(m)
p1	1308320.595	850864.926	1624.723
p2	1308318.008	850872.257	1624.898
p3	1308317.297	850881.755	1624.987
p4	1308311.896	850883.178	1625.185
p5	1308307.967	850887.295	1625.288
p6	1308304.292	850892.345	1625.333
p7	1308300.711	850894.386	1625.435
p8	1308299.569	850903.498	1625.423
p9	1308294.556	850901.156	1625.431
p10	1308291.566	850902.957	1625.401
p11	1308293.192	850914.432	1625.276
p12	1308291.304	850914.636	1625.182
p13	1308286.254	850918.129	1625.037
p14	1308282.538	850919.362	1625.004
p15	1308283.198	850928.139	1624.818
p16	1308279.587	850929.719	1624.693
p17	1308275.935	850930.596	1624.615
p18	1308274.926	850934.422	1624.544
p19	1308275.075	850938.721	1624.388
p20	1308271.780	850938.755	1624.323
p21	1308272.336	850944.340	1624.156
p22	1308267.409	850945.819	1624.039
p23	1308269.342	850953.337	1623.975
p24	1308267.339	850956.018	1623.774
p25	1308268.685	850959.772	1624.391
p26	1308262.943	850960.840	1623.559

p27	1308261.953	850965.250	1623.346
p28	1308263.837	850969.729	1623.405
p29	1308262.772	850975.312	1623.655
p30	1308258.782	850977.133	1623.077
p31	1308259.822	850982.078	1623.338
p32	1308263.053	850987.748	1622.391
p33	1308261.656	850992.766	1622.314
p34	1308258.983	850995.007	1622.294
p35	1308261.960	850996.162	1621.997
p36	1308262.455	850998.654	1622.155
p37	1308266.694	851004.888	1621.606
p38	1308265.619	851007.569	1621.619
p39	1308261.399	851009.701	1621.481
p40	1308258.374	851008.974	1621.603
p41	1308253.463	851008.996	1621.691
p42	1308250.931	851005.688	1621.760
p43	1308247.690	851006.508	1621.888
p44	1308243.783	851007.420	1622.085
p45	1308241.230	851004.759	1621.986
p46	1308236.047	851002.530	1622.157
p47	1308232.185	850998.478	1622.215
p48	1308228.783	850998.315	1622.283
p49	1308223.729	850999.656	1622.522
p50	1308221.247	850998.304	1622.809
p51	1308218.213	850996.437	1622.859
p52	1308214.656	850994.964	1622.918
p53	1308211.085	850992.843	1623.021
p54	1308207.868	850991.778	1622.501

p55	1308203.428	850992.200	1621.933
p56	1308201.225	850990.275	1621.927
p57	1308203.244	850986.478	1622.553
p58	1308201.286	850984.287	1622.673
p59	1308199.833	850981.963	1622.751
p60	1308199.825	850981.965	1622.752
p61	1308199.949	850978.406	1622.738
p62	1308201.135	850977.026	1622.668
p63	1308200.230	850972.913	1622.401
p64	1308202.407	850971.437	1622.780
p65	1308205.740	850969.360	1622.484
p66	1308207.525	850967.054	1622.515
p67	1308208.291	850963.369	1622.671
p68	1308206.476	850960.260	1622.740
p69	1308204.974	850957.616	1622.851
p70	1308203.081	850955.273	1622.920
p71	1308201.125	850955.743	1622.997
p72	1308199.608	850957.830	1623.110
p73	1308197.569	850957.950	1622.952
p74	1308196.560	850956.851	1622.990
p75	1308196.386	850955.508	1623.109
p76	1308195.876	850953.027	1623.448
p77	1308196.823	850951.005	1623.182
p78	1308199.025	850948.584	1623.070
p79	1308198.526	850946.295	1623.187
p80	1308196.817	850943.943	1623.265
p81	1308194.407	850941.267	1623.359
p82	1308190.472	850943.790	1623.299

p83	1308188.368	850943.133	1623.524
p84	1308185.957	850942.144	1623.657
p85	1308187.493	850939.568	1623.838
p86	1308190.309	850938.034	1623.457
p87	1308189.776	850935.251	1623.530
p88	1308187.823	850932.028	1623.605
p89	1308186.390	850929.579	1623.686
p90	1308184.113	850926.456	1623.767
p91	1308181.727	850923.417	1623.839
p92	1308178.256	850920.339	1623.900
p93	1308175.388	850917.035	1623.882
p94	1308171.399	850912.623	1623.925
p95	1308168.338	850909.014	1623.914
p96	1308165.336	850906.219	1623.901
p97	1308163.130	850903.449	1623.902
p98	1308158.655	850899.879	1623.916
p99	1308157.761	850897.010	1623.890
p100	1308155.521	850894.157	1623.884
p101	1308153.437	850891.668	1623.905
p102	1308151.895	850888.817	1623.889
p103	1308150.546	850886.610	1623.921
p104	1308147.840	850881.994	1623.900
p105	1308145.321	850878.366	1623.904
p106	1308141.428	850877.747	1623.703
p107	1308138.564	850874.893	1623.636
p108	1308136.163	850872.839	1623.586
p109	1308133.913	850871.672	1623.503
p110	1308131.158	850870.984	1623.359

p111	1308131.541	850868.777	1623.407
p112	1308129.785	850865.837	1623.354
p113	1308127.329	850863.162	1623.291
p114	1308125.369	850861.763	1623.183
p115	1308123.381	850861.665	1623.120
p116	1308123.111	850858.644	1623.084
p117	1308121.070	850856.263	1623.009
p118	1308119.674	850853.407	1623.019
p119	1308117.960	850850.621	1622.906
p120	1308115.018	850848.323	1622.753
p121	1308113.737	850845.471	1622.689
p122	1308112.003	850843.229	1622.649
p123	1308108.973	850844.490	1622.463
p124	1308109.199	850840.775	1622.478
p125	1308107.567	850837.837	1622.411
p126	1308105.696	850836.241	1622.348
p127	1308105.431	850833.883	1622.315
p128	1308104.404	850831.687	1622.260
p129	1308105.905	850828.555	1622.375
p130	1308102.649	850826.626	1622.224
p131	1308098.573	850827.654	1621.966
p132	1308093.934	850829.427	1621.576
p133	1308090.919	850830.618	1621.431
p134	1308088.448	850829.242	1621.293
p135	1308086.526	850830.040	1621.173
p136	1308083.592	850831.508	1620.999
p137	1308079.419	850828.741	1621.088
p138	1308074.666	850827.289	1620.562

p139	1308069.995	850824.865	1620.255
p140	1308066.392	850823.621	1620.066
p141	1308063.463	850821.092	1619.792
p142	1308059.142	850821.447	1619.099
p143	1308054.681	850824.181	1618.732
p144	1308051.839	850822.940	1618.643
p145	1308049.239	850818.400	1618.916
p146	1308047.800	850815.482	1618.854
p147	1308048.726	850813.063	1618.871
p148	1308051.613	850808.416	1619.222
p149	1308049.513	850805.976	1618.986
p150	1308050.340	850800.618	1618.739
p151	1308051.605	850797.931	1618.730
p152	1308054.194	850796.982	1618.979
p153	1308055.869	850794.858	1618.781
p154	1308055.521	850790.485	1618.622
p155	1308057.544	850787.423	1618.226
p156	1308059.616	850786.558	1618.273
p157	1308061.487	850786.608	1618.222
p158	1308062.166	850789.190	1618.380
p159	1308061.935	850792.434	1618.507
p160	1308062.912	850796.080	1618.672
p161	1308065.516	850800.301	1618.961
p162	1308070.584	850799.786	1619.723
p163	1308073.175	850797.896	1619.885
p164	1308075.574	850796.316	1619.973
p165	1308078.650	850793.935	1620.097
p166	1308085.279	850789.066	1620.283

p167	1308090.193	850785.467	1620.449
p168	1308096.222	850781.015	1620.604
p169	1308106.292	850772.645	1620.999
p170	1308115.639	850765.526	1621.215
p171	1308120.194	850761.726	1621.493
p172	1308124.412	850758.429	1621.614
p173	1308127.946	850755.735	1621.758
p174	1308135.138	850749.852	1622.082
p175	1308142.310	850743.242	1622.345
p176	1308142.310	850743.244	1622.345
p177	1308145.574	850739.835	1622.482
p178	1308154.370	850729.185	1622.727
p179	1308161.914	850719.803	1622.958
p180	1308166.043	850714.067	1622.985
p181	1308174.806	850700.255	1622.715
p182	1308177.551	850693.474	1622.541
p183	1308177.731	850693.026	1622.518
p184	1308178.779	850687.834	1622.437
p185	1308180.110	850677.164	1622.189
p186	1308181.055	850658.986	1622.037
p187	1308180.535	850653.571	1621.830
p188	1308177.881	850648.104	1621.447
p189	1308172.718	850644.718	1620.796
p190	1308168.256	850637.918	1620.014
p191	1308164.597	850611.168	1618.965
p192	1308169.528	850603.347	1619.023
p193	1308185.119	850611.674	1619.763
p194	1308192.860	850611.535	1620.052

p195	1308197.518	850615.154	1620.645
p196	1308198.428	850620.151	1620.713
p197	1308203.251	850622.766	1621.543
p198	1308204.180	850627.195	1621.571
p199	1308209.022	850632.682	1622.062
p200	1308215.119	850638.292	1622.473
p201	1308221.264	850633.972	1622.466
p202	1308224.746	850638.238	1622.678
p203	1308222.147	850647.180	1622.892
p204	1308231.313	850653.620	1623.318
p205	1308242.138	850655.937	1623.475
p206	1308231.915	850663.407	1623.525
p207	1308234.908	850671.502	1623.807
p208	1308241.293	850680.818	1624.069
p209	1308244.740	850685.488	1624.205
p210	1308250.628	850688.437	1624.281
p211	1308252.508	850692.778	1624.405
p212	1308256.718	850699.862	1624.546
p213	1308262.351	850708.053	1624.658
p214	1308267.336	850712.816	1624.825
p215	1308272.134	850716.213	1624.705
p216	1308272.560	850720.824	1624.849
p217	1308277.195	850725.300	1624.856
p218	1308282.258	850728.087	1624.796
p219	1308285.046	850732.410	1624.795
p220	1308285.238	850736.809	1624.908
p221	1308291.219	850738.876	1624.769
p222	1308293.274	850740.546	1624.795

p223	1308295.042	850742.507	1624.751
p224	1308295.832	850746.961	1624.770
p225	1308299.292	850748.835	1624.739
p226	1308301.909	850752.502	1624.715
p227	1308305.844	850755.640	1624.558
p228	1308310.251	850758.181	1624.468
p229	1308310.136	850762.181	1624.472
p230	1308313.922	850761.067	1624.302
p231	1308316.170	850764.880	1624.229
p232	1308316.945	850767.961	1624.226
p233	1308315.306	850769.922	1624.366
p234	1308313.142	850770.403	1624.489
p235	1308312.358	850775.269	1624.618
p236	1308311.458	850779.357	1624.619
p237	1308308.876	850782.465	1624.806
p238	1308306.466	850784.833	1624.894
p239	1308306.991	850787.769	1624.854
p240	1308307.639	850791.002	1624.809
p241	1308311.876	850793.965	1624.770
p242	1308311.415	850797.592	1624.718
p243	1308308.910	850800.394	1624.731
p244	1308305.138	850803.829	1624.672
p245	1308301.679	850808.256	1624.631
p246	1308302.289	850813.965	1624.608
p247	1308303.417	850820.615	1624.540
p248	1308308.831	850822.647	1623.842
p249	1308316.222	850823.784	1623.338
p250	1308320.845	850828.053	1622.562

p251	1308326.277	850830.193	1622.088
p252	1308330.993	850832.855	1621.774
p253	1308335.206	850838.374	1622.082
p254	1308338.194	850841.084	1621.847
p255	1308338.715	850844.261	1621.991
p256	1308337.170	850846.768	1622.417
p257	1308334.667	850848.062	1622.345
p258	1308333.659	850845.962	1622.598
p259	1308331.467	850846.653	1622.868
p260	1308328.186	850848.896	1623.481
p261	1308326.097	850849.935	1623.756
p262	1308324.024	850850.495	1623.970
p263	1308322.655	850851.848	1624.151
p264	1308324.903	850855.448	1624.097
p265	1308326.565	850857.238	1624.243
p266	1308324.745	850858.251	1624.236
p267	1308321.963	850858.528	1624.441
p268	1308321.404	850860.646	1624.601
p269	1308322.634	850863.036	1624.567
p270	1308324.320	850865.796	1624.425

Calculations

The area of the irregular field is calculated by the DGPS processing software using the corrected latitude and longitude coordinates of the boundary points.

Result

The area of the given irregular field was successfully and accurately determined using DGPS survey.

EXPERIMENT 7

Setting Out of Column and Foundation Using DGPS

Aim

To set out the column center points and foundation layout accurately on the ground using Differential Global Positioning System (DGPS) based on given design coordinates.

Apparatus / Equipment

DGPS rover receiver, controller/data logger, tripod, batteries, wooden pegs, hammer, lime powder or chalk, field notebook.

Detailed Procedure

The work is commenced by studying the foundation and column layout drawings to understand the location, spacing, and orientation of column center points. The design coordinates of all column centers and foundation corner points are prepared in advance in a suitable digital format compatible with the DGPS controller.

A DGPS base station is established at a known reference point with accurately known coordinates. The base receiver is mounted securely on a tripod, centered precisely over the reference point, and the antenna height is measured and recorded. The base station is powered on and allowed to initialize until it begins transmitting correction signals.

The DGPS rover receiver is then switched on and connected to the controller. The rover is allowed to initialize and receive real-time corrections from the base station. Once sufficient satellites are locked and acceptable positional accuracy is achieved, the rover is set to stake-out or navigation mode.

The first design point corresponding to a column center is selected in the controller. The rover is moved gradually on the site as guided by the controller display until the exact design coordinate is reached within the specified tolerance. When the correct position is obtained, the column center point is marked on the ground using a wooden peg.

The same procedure is repeated for all remaining column center points and foundation corner points. Each marked point is labeled clearly to avoid confusion. After marking all points, distances between adjacent columns and diagonals across the foundation layout are measured and checked against the design values to confirm accuracy.

Finally, lime powder or chalk is used to highlight the foundation lines by joining the marked points, and the entire layout is rechecked to ensure that the foundation and column positions are correctly set out as per the design drawing.

Tabulation

POINT NO	LATITUDE	LONGITUDE
A1	1308265	850840
A2	1308265	850825
A3	1308265	850818
A4	1308265	850788
B1	1308284	850840
B2	1308284	850825
B3	1308284	850818
B4	1308284	850809
B5	1308284	850788
C1	1308302	850840
C2	1308302	850809
C3	1308302	850788

Calculations

Coordinate differences between design points and field-marked points are displayed and verified directly using the DGPS controller. Linear distance checks are used to confirm layout accuracy.

Result

The column center points and foundation layout were successfully and accurately set out on the ground using DGPS.

EXPERIMENT 8

Measurement of Unknown Field Points and Stake-Out Using Total Station

Aim

To determine the coordinates of unknown field points and to stake out known design points accurately on the ground using a total station.

Apparatus / Equipment

Total station, tripod stand, prism with prism pole, measuring tape, wooden pegs, hammer, lime powder or chalk, field notebook.

Detailed Procedure

The survey work is started by selecting suitable control points with known coordinates in the survey area. The total station is set up firmly on the tripod over one of the known control points. The instrument is accurately centered using the optical or laser plummet and carefully leveled by adjusting the foot screws until the bubble is centered in all directions.

The total station is oriented by taking a backsight observation to another known reference point so that the instrument is aligned with the site coordinate system. After orientation, the prism mounted on a prism pole is placed vertically over the first unknown field point. The horizontal angle, vertical angle, and slope distance to the prism are observed. Based on these observations, the total station automatically computes and stores the Easting, Northing, and elevation of the unknown point.

The prism is moved sequentially to all other unknown field points, and their coordinates are measured and recorded in a similar manner. Care is taken to ensure that the prism pole is held vertical and that observations are stable before recording each point.

After determining the coordinates of the unknown field points, the design coordinates of the required stake-out points are entered into the total station or data collector. The instrument is then set to stake-out mode. The prism is moved on the ground as guided by the total station display until the exact design location is reached. Once the point is located within acceptable tolerance, it is marked on the ground using a wooden peg.

This procedure is repeated for all design stake-out points. After staking out all points, the positions are checked by measuring distances and angles between points to ensure accuracy. Lime powder or chalk is used to clearly mark the staked points on the ground.

Tabulation

A. Measurement of Unknown Points

Point No	Easting(m)	Northing(m)
1	0.000	0.000
2	1023.624	100.483
3	1020.433	100.476
4	963.385	98.851
5	968.076	98.766
6	970.921	99.088
7	970.749	98.959
8	972.825	99.108
9	977.147	99.276
10	986.117	99.461
11	994.125	99.621
12	1002.304	99.738
13	1012.072	99.755
14	1019.94	99.726
15	1028.369	99.535
16	1035.562	99.333
17	1043.615	99.04
18	1051.31	98.748
19	1056.544	98.528
20	1069.801	98.005
21	1076.746	97.64
22	1090.543	97.107

23	1103.889	96.568
24	1114.26	96.301
25	1121.567	96.54
26	1127.165	96.766
27	1126.94	96.904
28	1131.717	96.937
29	1134.849	97.011
30	1140.275	96.287
31	1137.241	96.863
32	1137.753	96.942
33	1137.694	96.94
34	1137.421	97.012
35	1135.567	97.025
36	1131.051	96.884
37	1126.904	96.77
38	1122.064	96.854
39	1120.437	96.974
40	1120.251	97.023
41	1120.268	97.174
42	1120.572	97.243
43	1120.49	97.302
44	1119.182	97.503
45	1119.171	97.621
46	1119.554	97.667

47	1124.438	97.82
48	1121.952	97.727
49	1120.569	97.822
50	1120.181	97.928
51	1120.148	97.978
52	1120.477	98.026
53	1121.702	98.191
54	1122.064	98.204
55	1122.842	98.192
56	1123.654	98.203
57	1124.204	98.205
58	1124.784	98.191
59	1125.519	98.19
60	1125.63	98.179
61	1126.305	98.18
62	1126.597	98.186
63	1126.872	98.165
64	1126.906	98.167
65	1126.476	98.192
66	1126.401	98.194
67	1126.306	98.191
68	1128.491	98.471
69	1128.19	98.493
70	1128.422	98.407

71	1129.604	98.318
72	1119.837	98.429
73	1108.115	98.399
74	1099.447	98.848
75	1093.903	99.135
76	1090.149	99.171
77	1086.094	99.274
78	1080.156	99.378
79	1076.991	99.462
80	1074.994	99.385
81	1070.533	99.632
82	1062.429	99.887
83	1054.755	100.077
84	1040.256	100.315
85	1033.593	100.317
86	1030.042	100.619
87	1026.125	100.354
88	1012.444	99.925
89	1007.983	99.864
90	1001.195	99.488
91	994.79	99.48
92	988.408	99.387
93	983.234	99.58
94	980.144	99.54

95	968.888	99.679
96	965.854	99.706
97	960.844	99.721
98	953.621	99.747
99	942.935	99.551
100	931.853	99.189
101	931.312	99.167
102	928.83	98.969
103	928.092	99.067
104	927.534	99.041
105	929.502	99.127
106	927.944	98.997
107	926.514	98.866
108	925.438	98.751
109	923.116	98.719
110	922.4	98.563
111	925.956	98.764
112	930.634	98.924
113	938.22	99.199
114	942.461	99.157
115	950.749	99.008
116	953.996	98.966
117	958.976	98.938
118	962.991	98.868

119	965.721	98.893
120	965.945	98.861
121	966.273	98.581
122	966.479	98.589
123	1023.624	100.483
124	1020.433	100.476
125	963.385	98.851
126	968.076	98.766
127	970.921	99.088
128	970.749	98.959
129	972.825	99.108

Stake-Out Points

Corner No	Design Easting(m)	Design Northing(m)
A1	1872.01	1695.18
A2	1872.01	1684.64
A3	1872.01	1675.21
A4	1872.01	1665.55
B1	1883.56	1695.18
B2	1883.56	1684.90
B3	1883.56	1675.21
C1	1891.42	1695.18
C2	1891.42	1684.64
C3	1891.42	1665.55

Calculations

Coordinates of unknown points are computed automatically by the total station using observed angles and distances. During stake-out, coordinate differences between design and field positions are displayed by the instrument and verified.

Result

The coordinates of the unknown field points were successfully determined, and the required design points were accurately staked out on the ground using the total station.

EXPERIMENT 9

Traditional Foundation Marking Using Rope and Peg Method

Aim

To set out and mark the foundation layout of a building on the ground using the traditional rope and peg method based on given plan dimensions.

Apparatus / Equipment

Measuring tape, rope or nylon string, wooden pegs, hammer, lime powder or chalk, spirit level, plumb bob, field notebook.

Detailed Procedure

The work is commenced by studying the building plan to understand the dimensions, orientation, and layout of the foundation. A suitable reference baseline is first established on the ground, preferably parallel to a permanent feature such as a road or compound wall. The baseline is marked using wooden pegs at both ends, and its alignment is checked carefully.

From the established baseline, the first corner point of the foundation is located using the measured distances as per the plan. A wooden peg is driven firmly into the ground at this point. Using the measuring tape, the lengths of the foundation sides are measured along the baseline and perpendicular to it.

Right angles at the foundation corners are established using the 3-4-5 method or by checking diagonal measurements. Once the right angle is confirmed, the remaining corner points of the foundation are marked using wooden pegs. The distances between opposite corners (diagonals) are measured and compared to ensure that the foundation layout is rectangular and accurate.

After confirming the correctness of all corner points, ropes are tied tightly between the pegs to represent the foundation lines. Lime powder or chalk is sprinkled along the ropes to clearly mark the foundation outline on the ground. The entire layout is rechecked for dimensions, squareness, and alignment before final approval.

Calculations

Right angle accuracy is verified using the 3-4-5 method and by checking equality of diagonals. Any minor corrections required are applied before final marking.

Result

The foundation layout of the building was successfully and accurately marked on the ground using the traditional rope and peg method.

EXPERIMENT 10

Interpretation of Total Station and DGPS Data and Plotting Using AutoCAD

Aim

To interpret survey data obtained from total station and DGPS and to prepare accurate drawings such as layouts, boundaries, and contours using AutoCAD software.

Apparatus / Equipment

Computer system, AutoCAD software, total station data files, DGPS data files, data transfer cable or USB device, field notebook.

Detailed Procedure

The survey data collected using the total station and DGPS during field work is first transferred to the computer using appropriate data cables or USB storage devices. The data files are checked to ensure that all required coordinate information such as Easting, Northing, and elevation are correctly recorded and free from errors.

The data is then converted into a format compatible with AutoCAD, such as text or CSV format. AutoCAD software is opened and a new drawing file is created. The drawing units are set appropriately based on the survey data. Using the coordinate input method, all survey points are plotted accurately in the drawing using their respective Easting and Northing values.

After plotting the points, they are joined in proper sequence using line or polyline commands to represent boundaries, layouts, or traverses. In the case of topographical surveys, spot levels are annotated and contour lines are generated by interpolation at the required contour interval. For layout and boundary surveys, building outlines, field boundaries, and reference lines are drawn clearly.

Once the drawing is completed, necessary annotations such as point numbers, levels, dimensions, north direction, scale, legend, and title block are added. The drawing is checked carefully for accuracy by comparing distances, areas, and orientations with the field data. Any discrepancies observed are corrected.

Finally, the completed drawing is saved in the required format and plotted or printed to the desired scale. The finalized drawing is documented as part of the survey record.

Calculations

Distance, area, and contour computations are carried out using AutoCAD commands based on the plotted survey data.

Result

The total station and DGPS survey data were successfully interpreted and accurate drawings were prepared using AutoCAD.